Microeconomics 2
The Consumer Problem
Instructor Notes

Arseniy Stolyarov

Updated 3 November 2026

Live route for Lecture 2

Everything before the board transition is click-by-click: budgets, existence, demand properties, indirect utility and their short proofs, followed by a budget-only Lagrangian and saddle-point introduction. The proof slides are in the main sequence rather than accessible only through appendix links. Pause at “To the board: derive Kuhn–Tucker” and derive the first-order system together. The later KKT slides are reference summaries; the worked examples, envelopes and Roy remain available afterwards. Expenditure minimisation and Slutsky belong to Lecture 3; the existing duality preview is reserve material.

The baseline is u continuous on $\mathbb{R}^L_+$, $p \gg 0$, $m > 0$ whenever existence or an attained value is used. LNS, convexity, strict convexity, smoothness and concavity are added only where needed. The continuity proof maps every budget to a fixed expenditure-share simplex and uses uniform continuity. No differentiability of demand or uniqueness is needed for that proof.

Two Lagrangians and the saddle argument

Before KKT, $\ell(x, \lambda) = u(x) + \lambda(m - p \cdot x)$ has domain $x \in \mathbb{R}^L_+$, $\lambda \geq 0$. Only the budget is relaxed. A saddle maximises in x and minimises in λ. Its right inequality gives budget feasibility and slackness; the left then proves optimality. This direction needs no concavity. The exact max–inf encoding is not a permission to exchange the order of optimisation. Later, the full $\mathcal{L}$ adds $\mu_i x_i$ to write first-order conditions for the quantity bounds explicitly. Under concavity, its KKT conditions verify the global saddle inequalities for the budget-only ℓ; the appendix proves this without evaluating utility at negative consumption.

Sources and course administration

Prepared from the user-supplied Lecture 10 (printed pp.149–165) and Lecture 11 (pp.166–180); author attribution was not verified in their metadata. Student reading remains Varian and optionally MWG. Original source-course homework notices are not imported. The current Homework 1 is already prepared for Lectures 1–2 and has

deadline 14 November 2026; use the latest homework folder rather than the old draft release wording.

Two theorems and their complete proofs

The live KKT section states and proves necessity and concave sufficiency, corresponding to the two theorems in source Lecture 11 p.171. The first proof constructs lambda and mu at any local optimum, proves common marginal returns and nonnegative signs, verifies budget and bound slackness, and covers the zero bundle. The second proves that the fixed-multiplier Lagrangian is concave and stationary, applies the supporting-plane inequality, and returns to utility through feasibility and slackness. The source p.164 algebra and stationary-concavity argument are steps of these complete proofs, not stand-alone substitutes for the theorems. The zero-gradient argument maximises only over a domain where utility is defined. The source's index and price typos are corrected.

Slide-by-slide notes

1. Microeconomics: The consumer problem

Briefly identify this as the next block, without repeating the entire course administration. Homework 1 is now issued after lecture block 3, not after this block.

2. From preferences to an actual choice

Keep the pre-KKT proof route on screen, then switch to the board at the marked transition. The later material is unchanged; duality and Slutsky are for Lecture 3.

3. The picture we already know

Recall the first lecture. Both prices are one and the budget is eight. The product cannot exceed sixteen, so sqrt(product) cannot exceed four. This example motivates a method but is not a universal tangency theorem.

4. The consumer problem

Use m consistently with lecture 1; the source handouts use w. We maintain strictly positive prices and wealth in the main KKT discussion. The case m=0 is a separate boundary case, with only zero consumption feasible.

5. What did we assume when we wrote that?

Do not claim that the consumer literally cannot save in any model: this static formulation omits saving unless it is included as a commodity. Subscriptions and quantity discounts require different feasible sets; the optimisation viewpoint still applies.

6. Two useful facts about the budget

Closedness follows from weak linear inequalities; boundedness uses every price being strictly positive. Scaling only prices, or only wealth, is not the same operation as changing the unit of account.

7. Proof: the budget is nonempty and compact

The upper bound uses both nonnegative quantities and strictly positive prices. The result is finite-dimensional Heine–Borel; no preference assumption enters this proof.

8. Does an optimum exist?

No concavity or monotonicity is needed for this existence statement. Keep existence, uniqueness and computation separate throughout. The term compact means closed and bounded here because the space is finite-dimensional.

9. Which assumption is doing the work?

For the last question use $u=x1+x2$ at $p=(1,1)$, any $m>0$: every budget-line bundle is optimal. The zero-price example is a failure example, not an assertion that all zero-price problems lack solutions.

10. Marshallian demand is initially a correspondence

Uppercase D is the correspondence and lowercase x the unique vector. This avoids using x(p,m) sometimes as a set and sometimes as a vector without warning. Marshallian demand is also called uncompensated demand.

11. Proof: demand is homogeneous of degree zero

The result is equality of correspondences, so no uniqueness assumption is necessary. A structural demand model should respect this restriction when all prices and nominal wealth are rescaled together.

12. Budget exhaustion: Walras' law

This is the consumer budget-exhaustion property often called Walras law. It is not a proof of existence. Saving is not ruled out if it is one of the goods. See the first lecture for satiation counterexamples.

13. Proof: LNS and budget exhaustion

The strict radius handles feasibility cleanly. Only an existing optimum is being considered. This proof is shared conceptually with lecture 1.

14. Proof: convex preferences give a convex demand set

This is an ordinal result using convex preferences, not concavity of the particular utility representation. It does not establish nonemptiness; the existence assumptions are maintained.

15. Strict convexity: at most one optimum

Answer to the last question: no. Strict convexity provides at most one optimiser on a convex feasible set. Continuity plus compactness supplied existence separately. Do not confuse strictly convex preferences with a strictly convex utility function.

16. Proof: uniqueness from strict convexity

No strict concavity of the chosen utility representation is required. For Cobb-Douglas on the closed orthant, establish positive optima before invoking strict properties on the positive orthant.

17. Indirect utility: the optimal value

v is a function even if Marshallian demand is a correspondence. It is ordinally scaled: changing the representation u changes v by the corresponding increasing transform.

18. Indirect utility: the properties to prove

The price statement is weak, not necessarily strict for a single price. LNS gives strict increase in wealth, but a derivative can still be zero at a point. Continuity follows from the maximum theorem; it is not differentiability.

19. Proof: currency units and higher prices

Budget inclusions are the whole argument. Hold the numerical representation u fixed. The price comparison is weak, and the vector inequality means all prices weakly increase.

20. Proof: more wealth and local nonsatiation

Use the same epsilon-ball argument as budget exhaustion, now spending the new slack to obtain a strictly better bundle. LNS supplies strictness. This does not assert a strictly positive derivative everywhere.

21. Continuity: put every budget on the same simplex

The expenditure-share simplex is a common compact set for every positive budget. Do not require exhaustion: the sum is at most one, not exactly one. No monotonicity, LNS or uniqueness is needed.

22. Proof: the optimal value is continuous

The maximum difference is bounded by the uniform difference of the objectives. Uniform continuity follows on a compact positive parameter neighbourhood times the simplex. Finite early sequence terms are irrelevant. This gives a complete short maximum-theorem argument in this particular budget problem.

23. A less obvious property: quasiconvexity

This is joint quasiconvexity in prices and wealth, distinct from quasiconcavity of direct utility in consumption. The source proof has priming typos; the same candidate bundle x must be used at both endpoint budgets.

24. Proof: indirect utility is quasiconvex

The identical bundle x must occur in both endpoint inequalities. The proof does not require convex preferences, concavity or LNS, only well-defined optimal values on the relevant budgets.

25. Before differentiating, inspect the economics

The source handouts emphasise not using the Lagrangian as a cookbook. Keep the method appropriate to the functional form. A nondifferentiable kink is not fixed by pretending that a derivative exists.

26. Let a multiplier enforce the budget

Use a separate symbol ell for the budget-only Lagrangian. Its x-domain remains the nonnegative orthant, so utility is never silently evaluated at negative consumption. The full L with mu appears later for the first-order system.

27. Why minimise in the multiplier?

The inner expression is an infimum, since infeasible bundles send the value to minus infinity. It is an exact extended-real encoding of the budget, not a minimax interchange theorem.

28. A global saddle point

This is now BEFORE KKT. Domain: x in the orthant, lambda nonnegative. The two inequality directions correspond to maximisation in quantities and minimisation in the budget multiplier. No claim that every arbitrary nonconcave optimum admits a saddle is being made.

29. Proof: the saddle enforces feasibility and slackness

Feasibility of quantities is imposed by the saddle domain, not derived by varying missing bound multipliers. Budget feasibility and its complementary slackness follow directly from minimisation over lambda.

30. Proof: a global saddle gives an optimal bundle

The three-inequality chain proves global optimality directly. Concavity is unnecessary for this direction; it matters when trying to establish the global maximisation inequality from local derivative conditions.

31. How could we find or verify a saddle?

Bridge to the board. Under concavity, KKT will verify the budget-only saddle; the appendix contains the short supporting-plane proof. Do not present arbitrary stationarity or a minimax interchange as sufficient.

32. To the board: derive Kuhn–Tucker

Pause the projection here and derive the KKT system at the board. The following slides are a ready summary, rather than a substitute for the derivation.

33. Why compare marginal utility per unit of money?

The feasible direction requires enough of good j to reduce it; this motivates why equality applies to consumed goods, while goods at zero require an inequality. The derivative calculation is local, not yet a global optimality proof.

34. The Lagrangian: one multiplier per inequality

The plus signs match the maximisation convention with constraints written as nonnegative. Do not mix with the minimisation convention. A log utility is not defined at zero: for that example establish an interior optimum first and work on the positive domain.

35. Stationarity in quantities

The notation u_i is partial derivative with respect to x_i. Because quantity bounds are represented explicitly, a corner does not mean deleting stationarity: its bound multiplier enters the equation.

36. The four Kuhn–Tucker groups

For p strictly positive and m>0, the active linear constraints satisfy the necessary constraint qualification. Differentiability on a neighbourhood is required for this form.

Necessity does not require concavity; global sufficiency does require additional structure.

37. Complementary slackness: read the products

No. A binding constraint may have zero multiplier. In particular, a budget can bind at a point where marginal value is zero. Appendix zero-shadow-value gives an explicit continuous LNS example. For substitutes at equal returns, an active zero-consumption bound may also have zero multiplier.

38. For consumed goods: equal marginal returns

Define MRS as u_i/u_j only where the denominator is positive and the indifference curve is differentiable. If all relevant marginal utilities vanish, division is invalid. Corners and kinks are handled differently.

39. For a good at zero: an inequality

It would force purchases of goods with too low a marginal return or generate negative quantities. The inequality is meaningful with positive prices. It is a necessary marginal condition, and under concavity part of a globally sufficient system.

40. A candidate is not yet a solution

The source has an interior-looking theorem followed by a broader constraint-qualification footnote. We state the smooth necessary result at corners too, with the qualification explained in the appendix. Concavity is a sufficient condition for KKT sufficiency, not a necessary property of preferences.

41. Theorem 1: KKT is necessary

This is the first explicit theorem on source Lecture 11 p.171, strengthened to include corners. The direct constructive proof supplies regularity for this particular linear budget; concavity is not used. We assume smoothness on a neighbourhood so gradients at the bounds make sense. Necessity holds even for a local optimum.

42. Construct the budget multiplier from an optimum

This is the construction in source Lecture 10 pp.162–164, corrected. Work at a nonzero optimum. Equal marginal returns for consumed goods follow from the preceding feasible-trade argument; nonnegativity follows by slightly reducing a consumed good. For the excluded zero-bundle case: if $x^* = 0$ is optimal and $m > 0$, a small increase in each coordinate is affordable, hence $u_i(0) \leq 0$. Set $\lambda^* = 0$ and $\mu_i^* = -u_i(0) \geq 0$; stationarity and complementary slackness hold. No concavity is needed for the sign argument.

43. Proof 1: consumed goods have one common return

For a pair of positive quantities, either sign of a sufficiently small spending transfer is feasible and preserves cost. If normalised marginal utilities differ, the transfer toward the larger one has positive directional derivative, contradicting even local optimality. This completes the multiplier-independence argument invoked on the construction slide.

44. Construct the quantity-bound multipliers

The equality $\mu_i^* = \lambda^* p_i - u_i(x^*)$ is the definition chosen to make stationarity true, not an implication that assumes dual feasibility. For consumed goods set mu to zero; the common-return result makes stationarity hold there too. The next two slides prove the missing sign.

45. If a constructed multiplier were negative

This is plain algebra and division by $p_i > 0$. In the source p.164 the left normalised derivative is mistyped: the consumed-good term must be u_j/p_j, not u_i/p_j. The same page later omits the price denominator when writing lambda for a consumed good. Both formulas here are corrected.

46. Proof: a negative multiplier permits a better purchase

The spending transfer is one-sided because the unconsumed good can be increased and the positive holding of j can be reduced. Positive directional derivative implies a strict improvement for all sufficiently small positive epsilon by differentiability; the finite utility change is not asserted to equal its linear approximation. This proves only $\mu_i \geq 0$, not strict positivity, including at a zero quantity.

47. Proof 1: finish with complementary slackness

This is the missing end of the source p.164 proof, completed on pp.165 and 169. Under budget slack a consumed coordinate can move in both directions, so its derivative is zero and lambda=0. Bound products are zero by construction, not by assuming the conclusion. The source p.165 has a reversed statement about which quantities make mu zero; the correct rule is positive quantity implies zero multiplier.

48. Proof 1: the zero-bundle case

This gives the separate source p.162 footnote in the main proof. Small positive coordinate increases are affordable since m>0. Their derivatives cannot be positive at a local optimum. Zero budget multiplier and minus the utility gradient supply all KKT conditions.

49. Theorem 2: concave KKT is sufficient

This is the second theorem on source Lecture 11 p.171. Concavity of u on the nonnegative orthant and differentiability near the candidate are sufficient. The source footnote gives a weakening using quasiconcavity PLUS a strict gradient implication. We use the simpler concavity theorem here. Quasiconcavity alone does not suffice, as the existing counterexample shows.

50. Fix the multipliers and look only at quantities

Fix the multipliers before discussing curvature in quantities. Affine terms preserve concavity, and constructed stationarity makes the gradient zero. Evaluate on a convex domain where u is defined and concave, for example the consumption orthant with a differentiable extension near the candidate. Do not silently extend log utility to negative quantities. The original source assertion for all real vectors needs a utility domain supporting it.

51. Proof: a stationary point of a concave function

The supporting-hyperplane inequality proves a global maximum, not just a local one. Concavity plus zero gradient is sufficient; strict concavity is needed only as a sufficient condition for uniqueness. A stationary point of a generic nonconcave function need not be optimal. The statement concerns the Lagrangian at fixed multipliers; feasibility and complementary slackness are still needed to connect it to consumer optimality.

52. A horizontal tangent above the whole graph

The first reveal shows the function, the second its tangent and candidate, and the third explains the word flat in the source. This is a one-dimensional illustration of the supporting-plane argument; the preceding slide is the multidimensional proof.

53. Proof 2: return from the Lagrangian to utility

The three inequalities prove consumer optimality from KKT without omitting feasibility or slackness. For the saddle interpretation we take the quantity domain to be the nonnegative orthant, where u is defined and concave. Full-L saddle minimisation follows from nonnegative slacks and zero products; no claim for negative quantities outside the domain is made.

54. Checkpoint

Answers: continuity on nonempty compact budget; strict convex preferences plus existence; nonnegative bound multipliers and slackness; concavity is a sufficient condition; a corner, infeasibility, a stationary nonmaximum or a nondifferentiable kink can be missed. No timed exercise label is used.

55. Example 1: Cobb–Douglas

Use the continuous raw Cobb-Douglas representation to establish existence and positivity on the compact nonnegative budget. Only then take logs, which are undefined at zero. The displayed exponent restriction is part of this two-good example.

56. Choose a convenient representation

The log transform preserves the positive-domain ranking. Do not apply the compactness theorem directly to the open positive budget; the previous raw-utility argument already supplied existence and excluded zero coordinates.

57. Solve the interior conditions

Concavity of the log objective and feasibility establish sufficiency. Positivity is satisfied because alpha, 1-alpha, prices and wealth are positive. The first-order equations are not being used alone without an optimality justification.

58. Interpret the solution

Constant expenditure shares and their invariance to prices/wealth are restrictions specific to Cobb-Douglas. Exhaustion here follows from preferences, while homogeneity follows from the budget construction. Do not present Cobb-Douglas shares as a fact about all consumers.

59. A price counterfactual

New x=(2,6), spending remains (6,6). Raw utility goes from sqrt(18) to sqrt(12). There is no meaningful additive utility contribution for each good in the multiplicative raw representation. The question is for students; the slide does not reveal the answers.

60. Example 2: perfect substitutes

An interior tangency equation need not have a solution when ratios differ, even though a global optimum certainly exists. This is why nonnegativity conditions matter.

61. Verify a corner with the full KKT system

All KKT groups hold, and the objective is affine/concave, so they certify global optimality. At the corner the subjective ratio 2 is not equal to the price ratio 3, and it should not be.

62. Equal returns: a whole set of demand

Preferences are convex but not strictly convex. A zero bound can be active with a zero multiplier at the endpoint, illustrating that binding does not imply positive multiplier.

63. Example 3: perfect complements

The argument is global and needs no derivative. For a more general Leontief min{x1/a,x2/b}, useful proportions are x1=a*t, x2=b*t and t=m/(a*p1+b*p2). An auxiliary-variable linear programme is a valid alternative, not differentiation of the kink.

64. Example 4: a boundary changes the regime

Reuse the lecture 1 example, now interpreting its KKT multipliers. y is positive whenever m>0, so its derivative is defined at the optimum. The square-root utility is not differentiable at y=0; that boundary was excluded economically. Smoothness is used locally around the candidate, while global sufficiency uses concavity on the consumption domain and its supporting-gradient inequality. At m=4, a=0 but its multiplier is zero.

65. A practical solution checklist

This is the main procedural summary for exercises. A formula with a negative quantity is not an acceptable answer to the nonnegative consumer problem. A smooth method is not a reason to discard nonsmooth optima.

66. From a classroom solution to a structural model

This connects to the lecturer’s requested motivation about real structural models. Identification of alpha is conditional on the maintained Cobb-Douglas model and complete measurement of the modelled budget. It is not an unconditional causal inference from one purchase.

67. Compute indirect utility in the same example

This v is the value of log utility, not the raw Cobb-Douglas value. Never combine its derivatives with the raw multiplier without accounting for the change of scale. The raw value is exp(v).

68. A continuous value need not be differentiable

Left derivative at p1=1 is -8, right derivative is zero. v is continuous but not differentiable. This example motivates the differentiability hypothesis in Roy’s identity. It is not a failure of existence.

69. The envelope idea

Present the envelope idea before derivative formulas. Do not mechanically differentiate a chosen x(p,m) and forget the binding constraints. The direct Lagrangian envelope includes the multiplier adjustment required by feasibility.

70. Differentiate the value, not the choice

Partial derivatives of the Lagrangian hold the choices and multipliers fixed while differentiating the parameters. Regularity is not automatic from continuity of u. These identities can also be established via supporting budgets; the appendix gives an elementary derivation under differentiable v.

71. The budget multiplier has an interpretation

For Cobb-Douglas log utility lambda=1/m. For the raw degree-one representation lambda=v_raw/m. Both give the same bundle. Strictly increasing v can have a zero derivative at a point, so positivity in Roy cannot be omitted.

72. Roy's identity: recover demand from value

The core has differentiable utility/regular constraints where envelope calculations are used. The more general direct budget-support proof in the appendix works with continuous LNS preferences and differentiable v. Positive v_m is essential, and ratio dimensions are quantities. The formula is not defined at the substitutes kink.

73. Check Roy in the Cobb–Douglas example

Use the same log utility scale throughout this calculation. This is an actual check against a separately derived demand, not only a restatement of the identity.

74. Demand survives a change of utility scale

The positive derivative condition is stronger than merely strictly increasing f. For instance a shifted cubic is strictly increasing but flat at its centre. Ordinal invariance of the optimisation problem remains true even if a particular differential formula degenerates.

75. Checkpoint

Answers: continuous u and nonempty compact budget for existence, strict convex preferences for uniqueness given existence; several maximisers share the same maximum value; marginal utility per price is no greater than lambda; concavity is a sufficient condition together with all KKT groups; differentiability of v and v_m>0 under the maintained consumer setup. Answers stay in the instructor notes.

76. Read next, and keep the core route clear

Assigned reading is the textbook. Use the already prepared Homework 1 on Lectures 1–2, deadline 14 November 2026; no old source-course HW5/HW6 notices.

77. Next block: two ways to ask the consumer question

This starts the optional block-3 preview from the tail of the supplied Lecture 11. It is not required before finishing the second-block core.

78. Expenditure minimisation

Capital H distinguishes a set of minimisers from a unique Hicksian vector h. A target above the attainable utility range makes the problem infeasible; do not state unconditional existence.

79. Why an unbounded feasible set can still have a minimum

The full upper contour set need not be bounded. Truncation by the cost of one feasible benchmark is enough. Continuity here refers to the selected utility representation.

80. Hicksian demand: basic properties

Targets are attainable, u continuous, p positive. If a nonzero minimiser exceeds the target, a slight proportional shrink remains feasible by continuity and is cheaper. If the target is u(0), zero is feasible and uniquely cheapest. For targets below u(0), no-excess utility need not hold.

81. The expenditure function has a different curvature

Holding the utility target fixed is essential. Concavity in prices does not require concave utility or convex preferences. Continuity on the attainable target domain is discussed under continuous utility and LNS in the instructor notes; avoid unqualified endpoint or infeasible-target claims.

82. Fixed wealth versus fixed utility

The red budget holds wealth fixed. The dashed compensated budget holds raw utility at four. Verify the compensated cost: 2*(2sqrt2)+4sqrt2=8sqrt2. The three points use different markers, not colour alone.

83. Two identities to develop next time

This is a preview, not a full Slutsky derivation. At target u(0), the expenditure is zero and v(p,0)=u(0); interpreting this endpoint extends the value function beyond the

positive-wealth core. More generally the corresponding solution sets agree under the maintained assumptions.

84. A monetary compensation question

CV is stated with the sign convention of extra wealth required after a price increase. The old budget is exhausted so e(p0,u0)=m. Monetary units do not automatically remove interpersonal or distributional normative choices; this qualifies an overly broad claim in the source.

85. Why the constraint qualification holds here

This is LICQ for the original linear budget and bounds with p positive and m positive. It does not hold automatically in every nonlinear constrained optimisation problem. If m=0 and all bounds and budget bind, the argument fails.

86. Proof: why concavity makes KKT sufficient

If lambda is zero the budget term is zero; otherwise slackness gives p*x*=m and p*y<=m. For bounds, mu*x*=0 and mu*y>=0. This is a direct global proof and does not need a separate saddle-point theorem.

87. Proof: concave KKT gives the budget saddle

This proves that concave KKT yields a global saddle for ell(x,lambda) on x>=0, lambda>=0. Only the budget is relaxed in ell; mu encodes the first-order bound conditions. Do not assert global maximisation of the full L over negative quantities where the utility may be undefined.

88. A KKT point that is not optimal

This shows why quasiconcavity alone is insufficient; the source’s additional strict gradient condition is not assumed in this counterexample. Pseudoconcavity or other additional conditions can restore sufficient first-order tests, but they are not needed in this core lecture. The example also shows that KKT does not by itself impose budget exhaustion at every candidate.

89. A binding budget with zero marginal value

Replace the raw utility by x and its multiplier is positive. The same ranking can have a flat differentiable representation at the optimum. This is why Roy states v_m>0 rather than only strict increase of v.

90. Cobb–Douglas with any positive exponent sum

The product representation need not be concave when the exponent sum exceeds one, but the log representation is concave on the positive orthant and shares the

same optimum. The source formula has a price-index typo; the denominator uses the price of good i, not always p1.

91. Roy without differentiating demand

This proof uses continuous LNS preferences, existence, a differentiable indirect value and a positive wealth derivative. It avoids assuming demand is differentiable or a priori single-valued, and does not require concave direct utility.

92. Proof: expenditure minimisation has a solution

The attainable-target assumption is indispensable. No convexity or LNS is required for this existence argument.

93. Proof: homogeneity and price concavity of expenditure

Monotonicity in prices follows by evaluating the higher-price minimiser at the lower prices. No excess utility follows by a small proportional contraction of any nonzero minimiser with surplus utility; this also supplies strict target monotonicity above baseline and uniqueness under strictly convex preferences.